

Touhidul Alam Seyam

MACHINE LEARNING ENGINEER | AI AND AGENTIC SYSTEMS

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RESEARCH PROFILE

Applied machine-learning researcher and software engineer working across computer vision, tabular modeling, healthcare analytics, agricultural AI, LLM inference, and high-performance computing. Author or co-author of ten source-backed research works, including two IEEE COMPAS papers, two SpringerOpen journal articles, and a 2026 Springer conference chapter. Builds reproducible Python pipelines and production-facing web systems that translate experiments into usable tools.

SELECTED WINS

- Champion, HackFusion InnoNation National Hackathon (solo Team Huntrix), 2026
- Champion, MicroOps Hackathon at CUET CSE IT Fest (with Abtahee Kabir), 2025
- Winner, RoboFusion 1.0 Techathon, 2026

RESEARCH AND ENGINEERING SKILLS

Machine Learning: scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CNNs, transfer learning, SVM, clustering, feature engineering, cross-validation, model evaluation

Engineering: Python, TypeScript, Next.js, React, Node.js, FastAPI, Django, PostgreSQL, MongoDB, Redis, Docker, Git, Zig, Bun

Research Practice: reproducible experiments, fold-safe validation, benchmark design, data pipelines, academic writing, literature review, publication preparation

ENGINEERING EXPERIENCE

Software Engineer | Agentic Institute | Remote | Dec 2025-Present

- Contribute to application development and AI-enabled, agentic software systems.
- Work across implementation, integration, testing, and delivery using modern web and backend technologies.

Software Engineer | Hello World Communications Ltd | Chattogram, Bangladesh | Aug 2024-Present

- Develop and deploy scalable applications and AI-enabled features using TypeScript, React, Next.js, Node.js, Python, and PostgreSQL.
- Bridge experimental software and production application requirements through code review, system design, deployment, and support.

SELECTED RESEARCH PROJECTS

bun-scikit | TypeScript, Bun, Zig | github.com/Seyamalam/bun-scikit

- Implemented native-accelerated classical ML with parity contracts, reproducible benchmarks, documentation coverage checks, and 209 tracked runtime exports.

Kaggriculture Autonomous Agent | Python, Simulation, Evaluation | github.com/Seyamalam/Kaggriculture

- Engineered a deterministic farming agent and evaluation harness with seeded seat-swapped tournaments, frozen replay corpora, manifest hashing, loss attribution, and regression gates.

Fold-safe ML Pipelines | Python, LightGBM, XGBoost, CatBoost | github.com/Seyamalam/playground-series-s6e8

- Built leakage-safe five-fold out-of-fold pipelines with nested target encoding, matched-fold screening, schema checks, rank blending, and a 0.96919 public ROC AUC submission.

Huntrix Delta | Expo, Go, gRPC, Offline AI | github.com/Seyamalam/hackfusion_huntrix

- Built autonomous triage and predictive route-decay features inside an offline-first disaster logistics system, supported by model-card methodology and a reproducible chaos simulator.

PUBLICATIONS

- “Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations.” Lecture Notes in Networks and Systems, 2026. DOI: [10.1007/978-3-032-15764-5_45](https://doi.org/10.1007/978-3-032-15764-5_45)
- “Enhancing Agricultural Diagnostics: Advanced Training of Pre-Trained CNN Models for Paddy Leaf Disease Detection.” Machine Learning Research, 2025. DOI: [10.11648/j.mlr.20251001.11](https://doi.org/10.11648/j.mlr.20251001.11)
- “Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data.” International Journal of Intelligent Information Systems, 2025. DOI: [10.11648/j.ijis.20251401.12](https://doi.org/10.11648/j.ijis.20251401.12)
- “Efficient Malware Classification Using Multiprocessing and Bag-of-Words Vectorization.” Advances in Networks, 2025. DOI: [10.11648/j.net.20251201.12](https://doi.org/10.11648/j.net.20251201.12)
- “Fine-tuning LLaMA 2 interference: a comparative study of language implementations for optimal efficiency.” arXiv, 2025. DOI: [10.48550/arXiv.2502.01651](https://doi.org/10.48550/arXiv.2502.01651)
- “AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management.” Journal of Electrical Systems and Information Technology, 2024. DOI: [10.1186/s43067-024-00169-7](https://doi.org/10.1186/s43067-024-00169-7)
- “Application of Machine Learning K-Means Clustering and Linear Regression in Determining the Risk Level of Pulmonary Tuberculosis.” IEEE COMPAS, 2024. DOI: [10.1109/COMPAS60761.2024.10796963](https://doi.org/10.1109/COMPAS60761.2024.10796963)
- “Enhancing Cardiovascular Risk Prediction Using Support Vector Machines and Advanced Machine Learning Algorithms.” IEEE COMPAS, 2024. DOI: [10.1109/COMPAS60761.2024.10796805](https://doi.org/10.1109/COMPAS60761.2024.10796805)
- “Comparing pre-trained models for efficient leaf disease detection: a study on custom CNN.” Journal of Electrical Systems and Information Technology, 2024. DOI: [10.1186/s43067-024-00137-1](https://doi.org/10.1186/s43067-024-00137-1)
- “Comparative Performance Evaluation of Classical Machine Learning and Quantum SVM for Heart Disease Prediction using a Quantum-Featured Dataset.” Sonargaon University Journal, 2025. Link: su.edu.bd/web_assets/journal/journal_five/journal3.pdf

EDUCATION

B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh | Chattogram, Bangladesh | Expected Dec 2026

Currently in the 8th semester. Research interests: applied machine learning, computer vision, agricultural AI, healthcare analytics, LLM systems, and agentic systems.

RESEARCH PROFILES

ORCID: 0009-0007-7512-1893 | Google Scholar: scholar.google.com/citations?user=gRkTVYEAAAAJ | GitHub: github.com/Seyamalam

LEADERSHIP AND ADDITIONAL HONORS

Assistant General Secretary | BGCTUB IT Club | Present - Help lead student technology activities, community programs, and club operations.

- 5th Place, 12th IUT ICT Fest GameJam, 2026; 5th Place, IIUC NextGen Hackathon, 2025
- Finalist, The Infinity AI BuildFest; Best Campus Ambassador, SciBlitz 2.0